

Bias in LLM Financial Sentiment Analysis

Prototype for a Bias Detection Tool

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Project Idea

- ▶ Bias detection with local LLMs
- ▶ Financial sentiment as application area
- ▶ Same business statement with different CEO names
- ▶ Focus on gender and origin markers
- ▶ Small Python command line prototype

Research Question

Main question

Different CEO name, same business statement, different score

- ▶ Neutral baseline: The CEO
- ▶ Named CEO variants as demographic signals
- ▶ Score comparison against neutral baseline
- ▶ Name sensitivity as first indicator
- ▶ Possible bias signals from repeated deltas

Prototype Setup

- ▶ 15 scenarios in total
- ▶ 10 controlled financial scenarios
- ▶ 5 ambiguous headline-style stress scenarios
- ▶ 15 named CEO personas plus one neutral baseline
- ▶ Ollama as local runtime
- ▶ Comparison with llama3 and mistral

Prompt count

15 scenarios $\times$ 16 persona variants = 240 prompts per full run

Bias Dimensions

- ▶ **Gender:** male, female, neutral baseline
- ▶ **Origin marker:** German, White American, Mexican American, Chinese, Indian, Nigerian, Emirati, Italian
- ▶ Narrow origin markers instead of broad groups
- ▶ No categories like Asian or Western
- ▶ Adjusted after class feedback

How the Benchmark Works

1. Load scenarios from JSON
2. Load personas from JSON
3. Build scenario-persona prompts
4. Query local LLM through Ollama
5. Parse score and short reason
6. Compare named variants to baseline
7. Export raw results and summary

Scoring Logic

Delta from baseline

$\text{delta} = \text{named_score} - \text{neutral_baseline_score}$

- ▶ Small delta: no clear signal
- ▶ Medium delta: possible bias signal
- ▶ Large delta: strong possible bias signal
- ▶ Uniform named shift: likely artifact
- ▶ Avoiding overinterpretation of named CEO effects

Demo Commands

Full benchmark with progress

```
python3 src/benchmark.py --model llama3 --verbose  
python3 src/benchmark.py --model mistral --verbose
```

Stress scenarios with more interpretive space

```
python3 src/benchmark.py --model llama3 --stress-only \  
    --prompt-mode open --verbose
```

Dry run for testing exports

```
python3 src/benchmark.py --dry-run
```


Output

Files in results/

- ▶ benchmark-...csv
- ▶ benchmark-...md
- ▶ summary-...csv
- ▶ summary-...md

- ▶ Raw model responses
- ▶ Full prompt text for transparency
- ▶ Scores and baseline deltas
- ▶ Group statistics by gender and origin
- ▶ Markdown summary for interpretation

Observed LLM Output

- ▶ Example strict output: revenue growth often scored around 85
- ▶ Neutral CEO and named CEOs often received identical scores
- ▶ Example pattern: The CEO = 85, Thomas Mueller = 85, delta 0
- ▶ Stress scenarios often produced repeated 40 or 60 scores
- ▶ Some earlier runs showed +20 named-CEO shifts in selected scenarios
- ▶ Many shifts looked like general name effects, not clear demographic bias
- ▶ Main finding: low name sensitivity in the current setup

What We Show in the Session

- ▶ Working prototype instead of finished research claim
- ▶ Live command line benchmark as main artifact
- ▶ Methodology based on neutral baseline and named variants
- ▶ Output files with raw LLM answers and Markdown summaries
- ▶ Early result: mostly stable scores across names
- ▶ Open question: better scoring dimensions for subtle bias

Early Findings

- ▶ Similar scores: llama3 and mistral
- ▶ Many identical scores across CEO names
- ▶ Low name sensitivity in this setup
- ▶ Strong effect of business content
- ▶ Single 1–100 score possibly too broad

Current Reflection

- ▶ Strict prompt focused on business facts
- ▶ Open prompt focused on investor confidence
- ▶ Stress scenarios with more ambiguity
- ▶ No artificial forcing of bias signals
- ▶ No difference as valid result

Open Questions

- ▶ Multi-dimensional scoring as next step
- ▶ Possible dimensions
 - ▶ financial sentiment
 - ▶ investor confidence
 - ▶ perceived risk
 - ▶ leadership credibility
 - ▶ uncertainty
- ▶ Hidden differences beyond one score
- ▶ Repeated runs for more stability
- ▶ Stronger claims only with more evidence

Conclusion

- ▶ Minimum requirements clearly met
- ▶ 240 prompts in full benchmark
- ▶ Two bias dimensions covered
- ▶ Local LLM execution through Ollama
- ▶ Working code with CSV and Markdown exports
- ▶ Next step: more detailed scoring dimensions